HEATSHIFT / HACKINSIGHT 26

Tomorrow, with evidence.

One model. Two preserved agents. A transparent temperature dashboard for everyday understanding.

The problem and our approach

People need understandable weather information and a way to see whether its supporting data can be trusted. HeatShift estimates airport temperature at the same hour tomorrow for Karachi, Lahore and Multan. It combines a frozen HistGradientBoosting model with Mateen's original Data Quality and Insight agents.

How it works

NOAA observations and archived weather inputs enter a documented warehouse. The quality agent fixes, fills or flags issues and records its decisions. The model uses a recent station reading, JMA forecast fields and calendar features. The Insight Agent retrieves trusted notes and calculates table statistics with row or section citations. Unsupported questions receive an evidence-limited answer.

Measured result	What it means
1.102 C model MAE vs 1.235 C baseline	1,783 reused validation rows; 10.8% lower average error. January was worse. Bootstrap interval includes no improvement.
March calibration: +/-2.22 / 3.08 / 2.27 C	Karachi / Lahore / Multan; nominal 90% bands. Future coverage is not guaranteed.
80.4 ms / 2,538 model rows	Warm local Node inference only; network, loading and browser startup excluded.

Delivery and impact

The responsive public Space contains predictions, data quality and cited insights. An hourly GitHub Actions workflow acquires live inputs, runs the preserved agents and frozen model, retains raw/cleaned/processed evidence and publishes status. Unavailable and stale connections are labelled. Potential benefit is clearer planning and data literacy; no user-impact study or paying customer has been established.

Limits and next steps

This is a three-airport temperature prototype, not a rain predictor or official warning service. Original JMA training-feature/search records and some original Open-Meteo collection metadata remain unavailable. No new locked-test, replay or reserve scoring was performed. Next: human review, recover provenance and independently test seasonal generalization.

Repository: https://github.com/waleedirfan2006-arch/Heat-Shift-
Live Space: https://huggingface.co/spaces/Waleed-1234/HeatShift-Dashboard

Team contributions

Waleed Irfan: developed and improved the prediction model. **Syed Mateen Ali:** developed, managed and maintained both AI agents: the Data Quality Agent and the Insight Agent. **Aman Khan Lodhi:** handled data cleaning, data warehouse management and related data preparation.

Roles supplied by Waleed Irfan. AI assistance is recorded separately in the cumulative AI use log.